

Title (30 words): Novel direct assessment of bulk fracture propagation speed in bone

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Abstract

Rib fractures are among the most common traumatic injuries, yet the mechanics of fracture propagation remain poorly quantified. Most current studies report only peak load or stiffness, overlooking crack development during the fracture. This study presents a validated, high-speed optical method to directly measure fracture propagation in rib bone. A custom three-point bending apparatus was developed, integrating a transparent acrylic impactor, dual high-intensity LED illumination, and a Phantom V711 camera. Bovine rib specimens (average cortical thickness ≈ 1.0 cm, span length ≈ 80 mm) were subjected to quasi-static loading until failure. Across eleven tests, six produced clear fracture propagation sequences capturing 3-6 frames of clear fracture footage corresponding to crack travel distances of 2-4 cm. Mean propagation velocity results were consistent with reported cortical bone fracture speeds. By providing quantitative parameters beyond peak failure load, this approach addresses a major limitation of current finite element injury models that treat fractures as an instantaneous event. Therefore, incorporating measured propagation velocities may improve the temporal accuracy of thoracic simulations, allow better predictions of rib fracture severity, and contribute to safety system design.

Keywords: rib fracture, biomechanics, fracture propagation, three-point bending, high-speed imaging, strain rate

Introduction

Bone fractures in traumatic events are common and often determine the extent of bony and soft tissue injury associated with acute and long-term morbidity and mortality (Rayner et al., 2025) [1]. However, the underlying biomechanics of bony fracture dynamics is still poorly understood (Derikx et al., 2011) [2]. For example, in the thorax, the rib cage protects the heart, lungs, major blood vessels and other mediastinal contents. The thoracic region often bears the dynamic brunt of both direct bony and soft tissue trauma when the body is exposed to automotive crashes or other blunt impacts (cf. Sharma, 2008) [3].

In current Lagrangian finite element modelling (FEM) of traumatic events, bone fractures are often characterized using a single peak stress or strain and their ultimate failure value (e.g. Mohammadi et al, 2026) [4]. Using these methods, the progression of bone fracture evolves by nonphysical reduction of load bearing either with heuristic local element deletion (Cui et al., 2022) [5] or cohesive zone degradation models that approximate failure through calibrated material parameters (e.g. Ural and Vashishth, 2006) [6]. Neither approach includes important details of how fractures initiate and evolve under potentially traumatic loads. Important factors not included in such modelling include the speed of crack propagation and complex

fracture paths as bone redistributes stress dynamically, fundamentally compromising the sub-element failure biomechanics for subsequent bony and soft tissue injury.

For example, rib fracture risk in human body thoracic finite element models is often calculated using strain based probabilistic frameworks without explicit fracture propagation kinetics (Forman et al., 2012) [7]. Other thoracic modelling approaches represent rib fracture through element deletion once local strain or failure thresholds are exceeded (Larsson et al., 2023; Zaseck et al., 2016) [8-9]. These methods can improve prediction of fracture risk, failure force, or fracture location, but they do not directly describe the time dependent motion of a fracture front through bone. This limitation is important because the redistribution of load after crack initiation depends on the rate and path of crack growth. Cohesive zone modelling provides a more physical alternative to simple element deletion, but its calibration and validation remain constrained by the limited availability of experimentally measured crack propagation speeds under dynamic loading (Ural et al., 2006) [6].

The speed of crack propagation in bone is expected to depend on applied loading rate, local micro and macrogeometry, stored elastic energy, and the dissipation/shielding mechanisms active during crack growth. At higher loading rates, stress and energy redistribution near a moving crack can strongly affect the subsequent structural response (Freund, 1990; Ritchie et al., 2005) [12-13]. Bone also differs from ideal brittle materials because it is an anisotropic heterogeneous biological material and thus its microstructure can produce crack deflection, uncracked ligament bridging, and rising crack growth resistance. These mechanisms create sensitivity to loading direction, rate, and environment, so crack speed cannot be inferred from peak load or quasistatic toughness measurements alone (Ritchie et al., 2005; Kulin et al., 2011; Zhai et al., 2020) [13, 10, 11].

Previous studies have provided important measurements of bone fracture resistance, but direct measurements of crack kinematics in bulk bone remain limited. Most fracture mechanics studies in bone have focused on local microfracture or quasistatic toughness measurements (Melvin, 1993) [14]. More recent dynamic studies using high speed synchrotron X-ray phase contrast imaging have measured notch initiated crack velocities of approximately 100-450 m/s in microscale cortical bone specimens under low velocity Kolsky bar loading (Zhai et al., 2019) [15]. Optical methods have also been used to track crack growth in small bone samples, including tracking of a visible whitening front when the crack itself is difficult to identify (Katsamenis et al., 2013) [16]. These studies demonstrate that crack motion can be measured but also show that the tracked feature must be clearly defined and that additional methods are needed for direct assessment of fracture propagation in bulk bone.

Since the structural response of dynamically fracturing bone depends on crack propagation speed, this study introduces a novel three point bone breaking apparatus for direct high speed optical tracking of fracture propagation in bulk rib bone. The principal hypothesis is that, following crack initiation, fracture propagation speed can be measured as a reproducible characteristic of the bulk bone material under the tested loading conditions. This study presents the design and pilot use of this system in dry bone and establishes a methodology for future investigations of bony fracture mechanics suited for incorporation into modern numerical models of traumatic injury.

Methodology

A custom three-point bend apparatus was constructed using a large-stroke, 10-ton hydraulic cylinder (ESCO ESC110) mounted in a rigid frame (Figure 1). The cylinder was mounted using a yoke to a transparent, acrylic indenter that provided access above the indenter for an illumination mount. The illumination mount provided a location for a high lumen light-emitting diode (LED) to backlight the bone through transparent indenter. The LED was a high-powered white light 30,000 lumen source (Citizen CLU05Q-3618E1-403L7X4) with cross sectional area of $\sim 960 \text{ mm}^2$ and a color temperature of 4000K.

Eleven dry bovine rib bones were acquired from a commercial supplier. Associated soft tissue and periosteum were cleaned from the bone specimen before testing. The outer margin of the rib specimens ($n=11$) were placed on cylindrical supports as shown in Figure 1. Fracture propagation was recorded using a high-speed camera (Vision Research, Phantom V711) at 210,000-390,000 frames/sec. Digital frames with the crack tip locations were tracked using Tracker Video Analysis and Modeling Tool (v.6.2.0). Fracture velocity was calculated through the slope of a linear regression fit to displacement and time data across the tracked frames. Differences between groups were assessed using two-sided Student's t-tests (Microsoft Excel for Mac, v. 16.78) with $\alpha=0.05$, $p=0.05$.

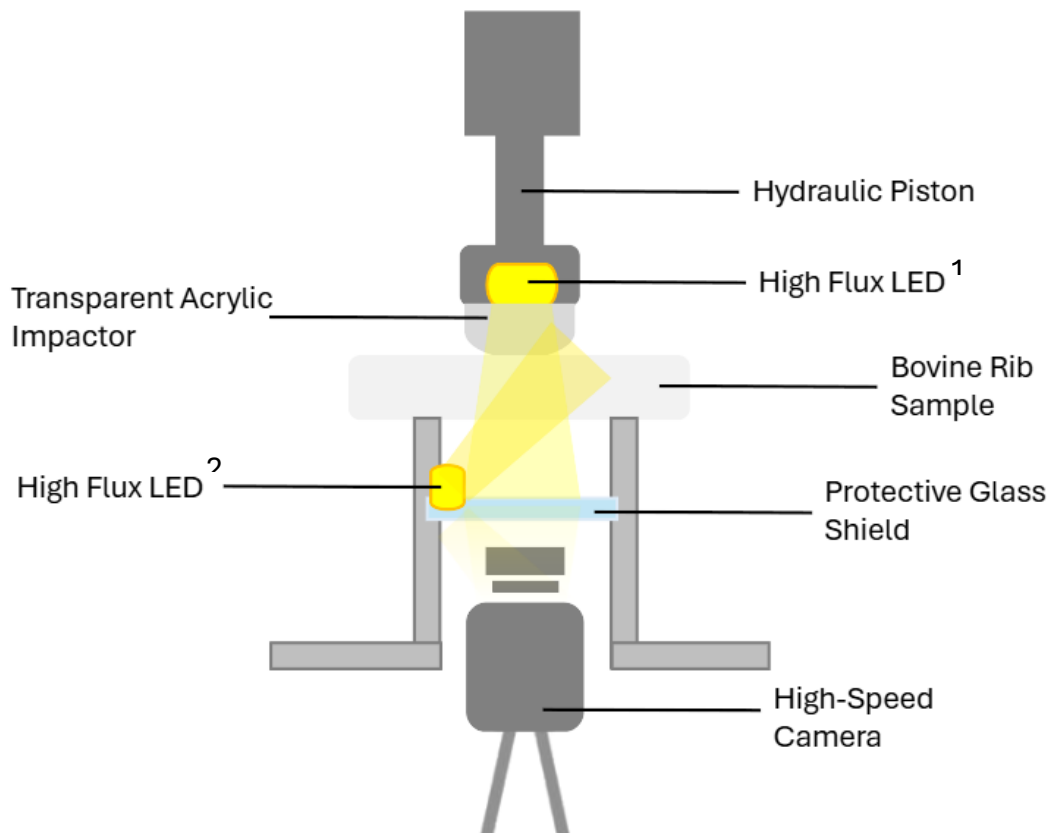


Figure 1. Novel three point bending apparatus to capture bone fracture propagation using a high-speed camera pointed under bovine rib sample. Two high flux LED modules illuminate the back-facing (1) and front facing (2) portions of the rib sample.

Results

Dynamic fracture tests were performed in three-point bend using low rate/high strain indentation with hydraulic cylinder loading. Crack propagation illuminated through bulk bone (Figure 2A) was successfully tracked for 3-6 frames in 11 tests (~400 $\mu\text{m}/\text{pixel}$ at 390,000 f/s). Typical fracture profiles were approximately linear during the time tracked with an average R^2 of $99\pm0.8\%$ (Figure 2B). Mean fracture velocity (mean $\pm$ standard deviation) was assessed as 1370 ± 200 m/s. There was no statistically significant difference in assessed velocities between the datasets with the smaller number of frames (3) and larger number of frames (4-6) ($\alpha=0.05$, $p=0.05$).

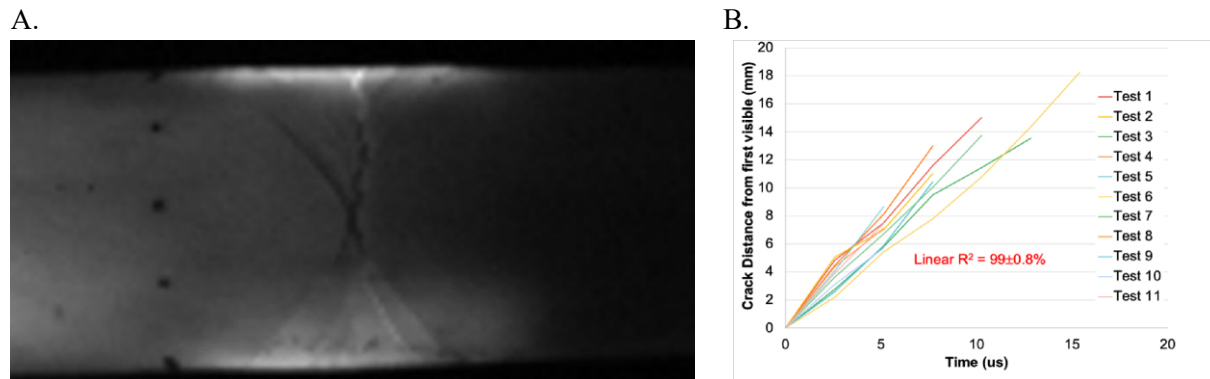


Figure 2. High speed tracking of bulk bone fracture propagation. A. Single video frame shows multiple cracks propagated across bone backlit through the transparent three-point bend indenter. B Video position tracking of crack position shows consistent linear behavior of crack propagation across bulk bone ($R^2 = 99\pm0.8\%$).

Discussion

Pilot tests on dry bovine ribs demonstrate that the apparatus designed for this study produces high contrast/high speed fracture recordings that are reproducible and are consistent in bulk bone. Pilot results show near-constant crack propagation speeds within specimens, consistent with expectations for cortical bone, though further work is needed to extend these results to hydrated bovine ribs and human cadaver samples.

The measured fracture velocities are consistent with theoretical predictions for limiting crack speeds in brittle solids. In traditional fracture mechanics, crack propagation in brittle materials is dictated by the Rayleigh surface wave speed, which represents the theoretical upper bound of crack tip velocity in mode I fractures (Freund, 1990) [12]. For cortical bone in the skull, the longitudinal wave speed has been measured using ultrasound methods as 2820 ± 40 m/s and the shear wave speed has been measured as approximately half of that of the longitudinal mode, 500 ± 140 m/s (White, 2006) [17]. The Rayleigh wave speed is frequently measured as 90-92% of the shear wave speed (Dong & Guo, 2004) [18]. The observed velocities in this study of approximately 1370 m/s fall within this physically plausible range for the Rayleigh speed

of dry cortical bone. The correspondence between the results of this study and the bony Rayleigh speed suggests that under sufficiently high loading rates, cortical bone fracture propagation approaches a limit dependent on bulk material properties alone, rather than the measured quasistatic values that are dependent on stress and intensity. This important observation implies that, above a certain threshold stress value and strain rate, the crack does not have a tendency to accelerate indefinitely but is bound by the elastic wave characteristics of the bone. This is behavior observed in glass, ceramic, and tough polymers, but rarely in hard biological material.

Fracture propagation speed is an essential input into biofidelic human FEMs that include fracture. The practical motivation for this work lies in improving the biofidelity of computational models used to predict traumatic injury. Contemporary Lagrangian finite element models of the thorax, including the widely used THUMS and GHBM, simulate bone failure through simply with element deletion when nodal stress and/or stress exceed a given threshold. This approach has two fundamental drawbacks. First, element deletion does not represent a physical process, rather it is a numerical convenience that instantaneously removes material and energy from the system. This effectively assigns infinite fracture propagation speed across the element at the moment of failure and failure propagation speeds that depend on details of previous element deletions and computational timesteps. Effectively, the size of the deleted element determines the effective spatial and temporal resolution of the simulated fracture, introducing a mesh-dependency that is well recognized but difficult to resolve without first establishing fracture laws within the experimental data (Cui et al., 2022) [5].

The crack propagation velocities measured here provide a concrete target that can be used to calibrate and validate biofidelic fracture models. Extended finite element methods (XFEM) and cohesive zone models (CZM), for example, can represent crack propagation as a physically continuous process with an explicit velocity, but their implementations in bone injury contexts is limited by the absence of experimentally validated propagation speeds under failure load rates (Ural et al., 2006) [6]. Crack propagation speed found in this study is substantially higher than velocities assumed or implied in most existing bone FEM studies. Integrating this category of data into FEM validation frameworks is expected to improve the temporal accuracy of fracture patterns and the redistribution of stresses and subsequent hard and soft tissue injury following crack initiation.

An important limitation of the present study is that all specimens were dried bovine ribs. Hydration state is known to significantly affect the mechanical behavior of cortical bone. Wet bone exhibits substantially greater fracture toughness and post-yield ductility compared to dry bone, largely due to the plasticizing effect of water on the collagen network and the viscoelastic contribution of bone fluid under dynamic loading (Nyman et al., 2006; Viguet-Carrin et al., 2006) [19,20]. Crack propagation in hydrated bone is expected to be slower, more winding, and associated with greater energy dissipation through crack bridging and microcracking ahead of the crack tip; all mechanisms that are suppressed in dry specimens. Therefore, the relationship between our observed velocity and the theoretical Rayleigh wave speed, may reflect the limiting behavior of the dry, brittle case rather than the biomechanically representative case.

This novel light-based backlighting methodology in this study provides a practical way to quantify fracture initiation, propagation speed, and terminal branching behavior. By filling this data gap, it supports more realistic human body trauma finite element models and opens new paths for improving injury biomechanics research and safety applications.

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